

since probab. "i" would have in degree = $K_b = p(K_b, t) = \frac{n(K_b, t)}{N_b(t)}$

$$(2) \Rightarrow [(1+1)p(K_b, t+1) - t p(K_b, t)] (1-g)$$

$$= \tilde{Z}^{-1} [(\lambda_b(K_b+1) + \mu_b) p(K_b+1, t) - (\lambda_b K_b + \mu_b) p(K_b, t)] (1-g) + (1-g) \delta_{K_b, 0}$$

$$\lim_{t \rightarrow \infty} p(K_b, t) = p_{\infty}(K_b) :$$

$$\Rightarrow \lim(2) : p_{\infty}(K_b) = \tilde{Z}^{-1} [(\lambda_b(K_b+1) + \mu_b) p_{\infty}(K_b+1) - (\lambda_b K_b + \mu_b) p_{\infty}(K_b)] + \delta_{K_b, 0} \quad (3)$$

$$\text{if } K_b > 0 \Rightarrow \delta_{K_b, 0} = 0$$

$$\Rightarrow p_{\infty}(K_b) [\tilde{Z}^{-1} (\lambda_b K_b + \mu_b) + 1] = p_{\infty}(K_b+1) [\tilde{Z}^{-1} (\lambda_b K_b + (\mu_b - \lambda_b))]$$

$$\Rightarrow \frac{p_{\infty}(K_b)}{p_{\infty}(K_b+1)} = \frac{\tilde{Z}^{-1} \lambda_b K + \tilde{Z}^{-1} \mu_b - \tilde{Z}^{-1} \lambda_b}{\tilde{Z}^{-1} \lambda_b K + \tilde{Z}^{-1} \mu_b + 1} = \frac{K_b + \mu_{\frac{1}{2}b} - 1}{K_b + \mu_{\frac{1}{2}b} + \frac{\tilde{Z}}{\lambda_b}}$$

$$\Rightarrow p_{\infty}(K_b) = A_b \cdot \frac{\Gamma(K + \mu_{\frac{1}{2}b})}{\Gamma(K + \mu_{\frac{1}{2}b} + (1 + \frac{\tilde{Z}}{\lambda_b}))}$$

$$\text{for } K_b \geq 0 : p_{\infty}(K_b=0) \Rightarrow (3) \rightarrow p_{\infty}(0) = \tilde{Z}^{-1} [-\mu_b p_{\infty}(0)] + \delta_{0,0}$$

$$\Rightarrow p_{\infty}(0) (1 + \mu_b \tilde{Z}^{-1}) = 1 \Rightarrow p_{\infty}(0) = (1 + \mu_b \tilde{Z}^{-1})^{-1}$$

using (3) from $K > 0 :$

$$\therefore p_{\infty}(0) = (1 + \mu_b \tilde{Z}^{-1})^{-1} = A_b \frac{\Gamma(\mu_{\frac{1}{2}b})}{\Gamma(\mu_{\frac{1}{2}b} + (1 + \frac{\tilde{Z}}{\lambda_b}))}$$

$$\Rightarrow A_b = \frac{\Gamma(\mu_{\frac{1}{2}b} + (1 + \frac{\tilde{Z}}{\lambda_b}))}{\Gamma(\mu_{\frac{1}{2}b})} \cdot (1 + \mu_b \tilde{Z}^{-1})$$

conjecture by problem symmetry $\Rightarrow$

$$p(K_a^{(n)}) = A_a \frac{\Gamma(\mu_{\frac{1}{2}a})}{\Gamma(\mu_{\frac{1}{2}a} + (1 + \frac{\tilde{Z}}{\lambda_a}))}$$

$$A_a = \frac{\Gamma(\mu_{\frac{1}{2}a} + (1 + \frac{\tilde{Z}}{\lambda_a}))}{\Gamma(\mu_{\frac{1}{2}a})} \cdot (1 + \mu_a \tilde{Z}^{-1})$$